Stokes fluid and shallow water modeling

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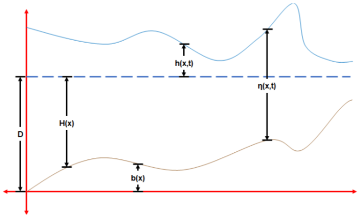
2D shallow water equation

$L \gg \eta$, η – total water depth, L – characteristic length scale.

Conservative form (no Coriolis, frictional or viscous forces)



$$\frac{\partial \eta}{\partial t} + \frac{\partial (\eta u)}{\partial x} + \frac{\partial (\eta v)}{\partial y} = 0$$

$$\frac{\partial (\eta u)}{\partial t} + \frac{\partial}{\partial x}\left(\eta u^2 + \frac{1}{2} g \eta^2\right) + \frac{\partial (\eta u v)}{\partial y} = 0$$

$$\frac{\partial (\eta v)}{\partial t} + \frac{\partial}{\partial y}\left(\eta v^2 + \frac{1}{2} g \eta^2\right) + \frac{\partial (\eta u v)}{\partial x} = 0$$

Non-conservative form

$$\frac{\partial h}{\partial t} + \frac{\partial}{\partial x}\left((H+h)u\right) + \frac{\partial}{\partial y}\left((H+h)v\right) = 0$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} - fv = -g\frac{\partial h}{\partial x} - ku + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + fu = -g\frac{\partial h}{\partial y} - kv + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$$

u – x-velocity, v – y-velocity, H – mean water depth, h – water height deviation from its mean, $\eta = H(x,y) + h(x,y,t)$, k – viscous drag coefficient, ν – kinematic viscosity, g – acceleration due to gravity, f – Coriolis coefficient.

SWE on sphere

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial \lambda} \cdot \frac{u}{a \cos\theta} + \frac{\partial u}{\partial \theta} \cdot \frac{v}{a} - \left(f + \frac{u}{a} \tan\theta\right) \cdot v + \frac{\partial \eta}{\partial \lambda} \cdot \frac{g}{\cos\theta} = 0$$

$$\frac{\partial v}{\partial t} + \frac{\partial v}{\partial \lambda} \cdot \frac{u}{a \cos\theta} + \frac{\partial v}{\partial \theta} \cdot \frac{v}{a} + \left(f + \frac{u}{a} \tan\theta\right) \cdot u + \frac{\partial \eta}{\partial \theta} \cdot \frac{g}{a} = 0$$

$$\frac{\partial h}{\partial t} + \frac{\partial h}{\partial \lambda} \cdot \frac{u}{a \cos\theta} + \frac{h}{a \cos\theta} \left(\frac{\partial u}{\partial \lambda} + \frac{\partial (v \cos\theta)}{\partial \theta}\right) = 0$$

u and v are respectively λ- and θ- (longitude and latitude) components of velocity in spherical coordinates, a - Earth radius.

$$(t, \lambda, \theta) \in \Omega = [0, t_f] \times [-\pi, \pi] \times [-\pi/2, \pi/2]$$

Spherical Fourier Neural Operators[1]

Spherical harmonics, SHT

$$Y_l^m(\theta,\lambda) = N_{lm} P_l^m(\sin\theta) e^{im\lambda} \text{ – Laplace-Beltrami eigenfunctions}$$

$$\textbf{SHT:}\ f(\lambda,\theta) = \sum_{l=0}^{\infty}\sum_{m=-l}^{l} Y_l^m(\theta,\lambda)\hat{f}_{lm}, \quad \hat{f}_{lm} = \iint_{\mathbb{S}} f(\lambda,\theta)\overline{Y_l^m(\theta,\lambda)}d\Omega$$

Model

Input: $x = (h(t), u(t), v(t)) \in \mathbb{R}^{3\times N_\theta \times N_\lambda}$

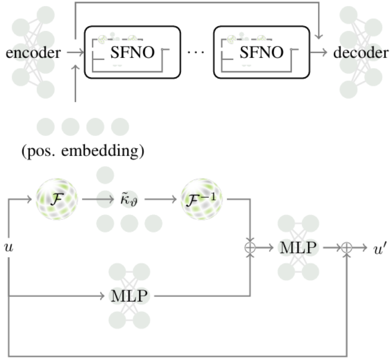
Output: $y = (h(t+\Delta t), u(t+\Delta t), v(t+\Delta t))$

$\tilde{\kappa}_\nu$ - set of linear layers per each $l = \overline{0, l_{max}}$

$\mathcal{F}, \mathcal{F}^{-1}$ - SHT and inverse SHT

$\hat{y} = \text{Decoder}\left((\text{SFNO\_block})^{(k)}(\text{Encoder}(x))\right)$

$$\textbf{Loss:}\ \mathcal{L}_\Theta = \frac{1}{N}\sum_{i=1}^{N}\frac{\|\hat{y}^{(i)} - y^{(i)}\|^2_{L^2(\mathbb{S})}}{\|y^{(i)}\|^2_{L^2(\mathbb{S})}} + \frac{\alpha_M}{N}\sum_{i=1}^{N}\left|\frac{M(\hat{y}^{(i)} - M(y^{(i)}))}{M(y^{(i)})}\right|$$

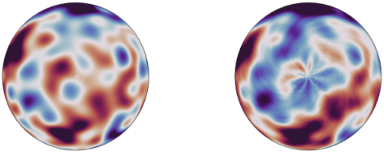


[1] https://arxiv.org/pdf/2306.03838

Motivation: Geometry and Equivariance

Problem: Standard FNO (FFT)

- FFT implies flat Torus topology $\mathbb{T}^2$ (periodic on x, y).
- $\mathbb{S}^2 \ncong \mathbb{T}^2$: Poles are singularities.
- **Artifacts:** «Ringing» and dissipation near poles (see Fig.).
- **No** $SO(3)$ **Equivariance:** Rotating Earth $\neq$ Translating grid.



SFNO vs FNO

Solution: Spherical Convolution

Define convolution via rotation $R \in SO(3)$ instead of translation:

$$(\kappa \star u)(x) = \int_{SO(3)} \kappa(Rn) \cdot u(R^{-1}x) dR$$

- $n = (0, 0, 1)^T$ (North pole).
- κ: learnable filter kernel.

Convolution Theorem on $\mathbb{S}^2$

Convolution becomes multiplication in Harmonic domain:

$$\mathcal{F}[\kappa \star u]_{lm} = 2\pi \sqrt{\frac{4\pi}{2l+1}} \cdot \hat{\kappa}_{l0} \cdot \hat{u}_{lm}$$

$$\implies \hat{y}_{lm} = \tilde{W}_l \cdot \hat{x}_{lm}$$

Result: Block-diagonal operations (mix channels, independent per l).

Stokes fluid

High and constant viscosity, Re $\ll 1$ (low Reynolds number), assuming incompressible Newtonian fluid:

Staionary equations (linear!)

$$\mu \nabla^2 \mathrm{u} - \nabla p + \mathrm{f} = 0$$

$$\nabla \mathrm{u} = 0$$

u – velocity, p pressure, μ – dynamic viscosity

Non-staionary equations

$$\rho \frac{\partial \mathrm{u}}{\partial t} + \mu \nabla^2 \mathrm{u} - \nabla p + \mathrm{f} = 0$$

$$\nabla \mathrm{u} = 0$$

Boundary conditions

«non-slip»: $\mathrm{u}(\mathrm{x}) = \mathrm{u}_{wall}, \quad \mathrm{x} \in \partial\Omega$

The Green's function recap

Definition: For a linear differential operator $\mathcal{L}$ and boundary constraint $\mathcal{D}$:

$$\begin{cases} \mathcal{L}G(x,y) = \delta(y-x), & x \in \Omega \\ \mathcal{D}G(x,y) = g, & x \in \partial\Omega \end{cases}$$

G – Green's function

The Superposition Principle: Instead of solving $\mathcal{L}u = f$ numerically for every new f, we compute:

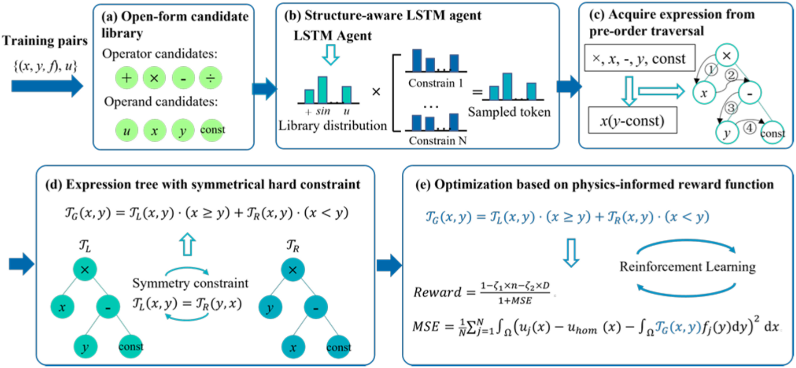
$$u(x) = \underbrace{\int_\Omega G(x,y)f(y)dy}_{\text{Particular Sol.}} + \underbrace{u_{hom}(x)}_{\text{Homogeneous Sol.}}$$

What for?

- **Universal:** Encodes the physics of the domain geometry.
- **Fast Inference:** New solutions require only integration, not PDE solving.

Discovery of Green's function based on symbolic regression with physical hard constraints[2]

Input: $\{(f_j, u_j)\}_{j=1}^N$, **output:** A symbolic formula $\mathcal{T}(x, y)$.
Inference (for forcing term f_{new}): $f_{new} \to \int \mathcal{T} f_{new} \to u_{new}$.
LSTM Agent $\longrightarrow$ sequence of tokens in the tree in order of DFS traversal
Vocabulary: $\{+, -, \times, \div, \sin, \exp, x, y, \xi_1, \xi_2 \dots\}$, variables in the leaves only
The actual numbers in the formula in place of constants optimized via BFGS



[2] https://arxiv.org/pdf/2408.00811

Optimization

$\mathcal{T}(x, y; \xi)$ – candidate function, u_{hom} – homogeneous solution
Step 1: Loss (Physics-Informed MSE) Calculated after BFGS tunes ξ for a fixed structure

$$\text{MSE} = \frac{1}{N}\sum_{j=1}^{N}\int_{\Omega}\left(u_j(x) - u_{hom}(x) - \int_{\Omega}\mathcal{T}(x, y; \xi) f_j(y) dy\right)^2 dx$$

Step 2: Reward

$$R = \frac{1 - \zeta_1 n - \zeta_2 D}{1 + \text{MSE}}$$

$\zeta_{1,2}$ are penalty factors for complexity, D - tree depth, n – formula length.
Step 3: Policy Update (Risk-Seeking Policy Gradient)
The LSTM parameters θ are updated to maximize the probability of high-reward formulas:

$$\nabla_\theta J(\theta) \approx \frac{1}{M}\sum_{k=1}^{M}(R_k - R_{\text{baseline}})\nabla_\theta \log P(\text{expression}_k|\theta)$$